Manual for Transmitting, Receiving, and Decoding CW and GFSK

Using RFM26W, SDR#, CWGet, SoundModem, and RTL-SDR Dongle

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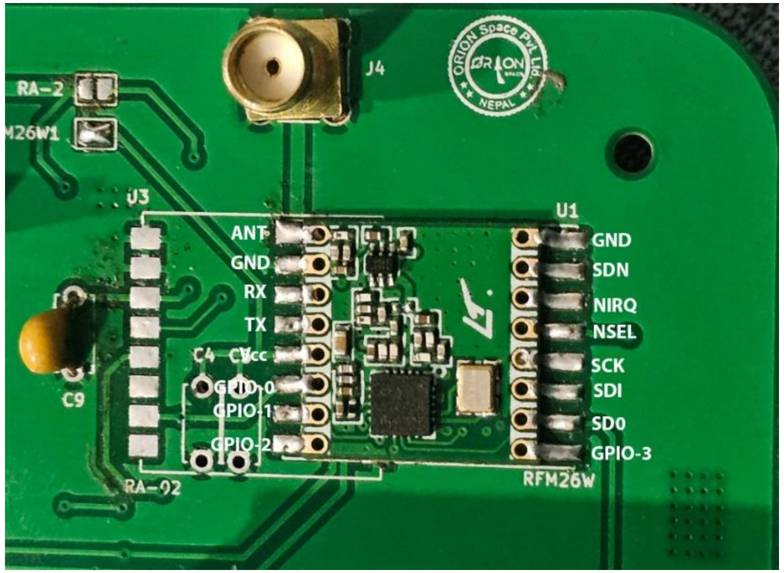
OBJECTIVE

Demonstrate how to:

- Transmit CW and GFSK signals using RFM26W
- Receive signals with an SDR dongle via SDR#
- Decode CW with CWGet
- Decode GFSK with SoundModem

RFM26W MODULE OVERVIEW

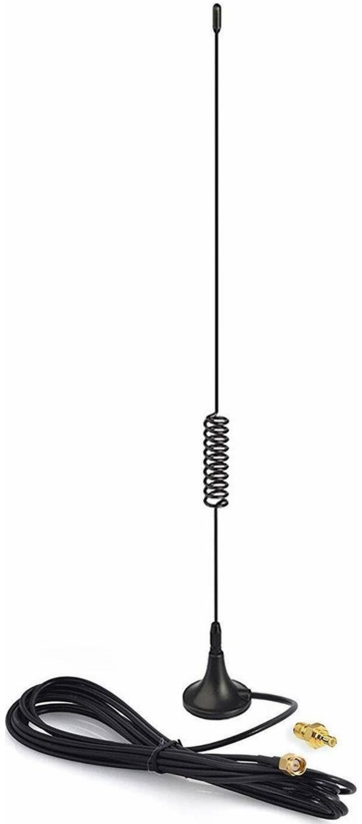
- High-performance sub-GHz transceiver (142–1050 MHz) with –126 dBm sensitivity
- Ultra-low current consumption in active and standby modes
- Excellent phase noise, blocking, and selectivity for narrowband applications
- PCB layout: ANT pin positioned at top-left, below antenna connector (J4) as shown in figure.



RFM26W on PCB

REQUIRED HARDWARE

RTL-SDR Dongle



Antenna

PC/Laptop

SOFTWARE REQUIRED

- Arduino IDE (for RFM26W programming)
- SDR# (SDRSharp) for signal reception
- CWGet for CW decoding (morse)
- SoundModem for GFSK decoding
- VB-Audio Cable for virtual audio routing through ports

LIBRARIES REQUIRED

- Mini Morse (.h and .c files)
- Radio Head (RH_RF24)

Note: Download and copy-paste these libraries at:

C:\Users\Documents\Arduino\libraries

SOFTWARE REQUIRED

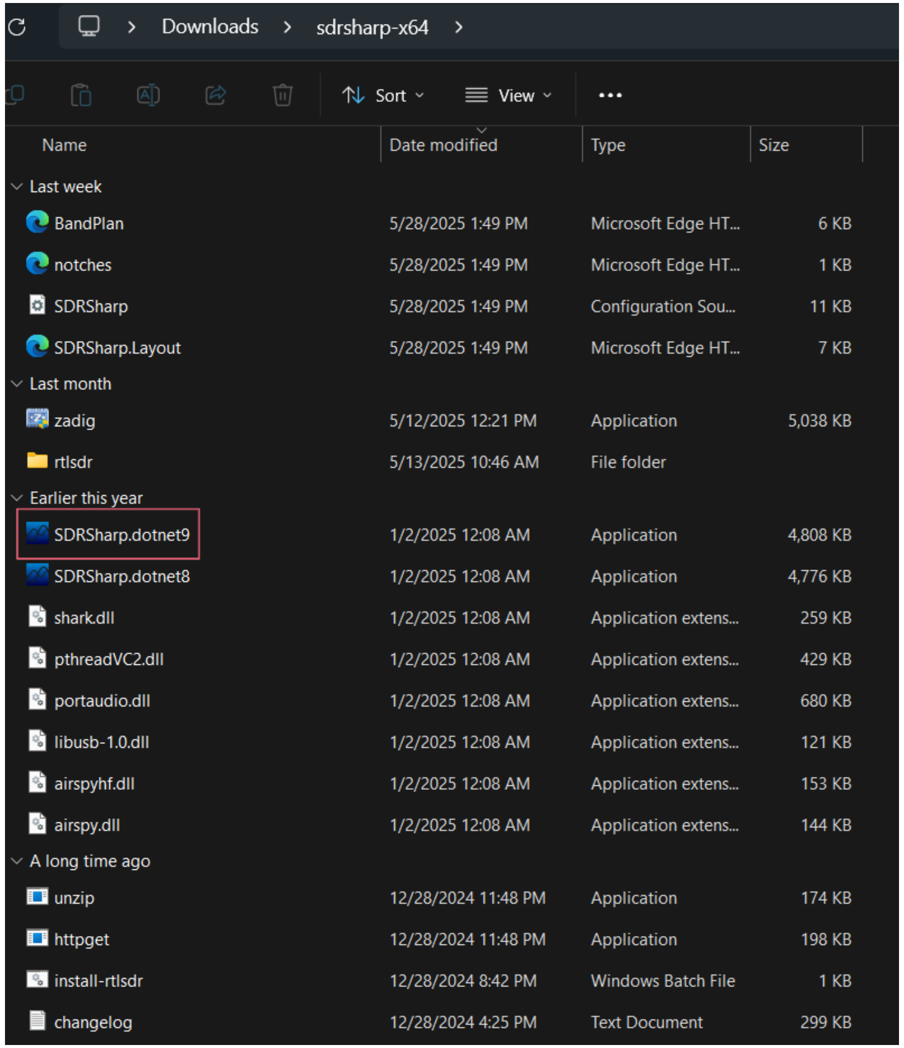
- AirSpy SDR# : SDR# and Airspy Downloads - AIRSPY
- CWGet: CW morse code decoder
- Sound Modem
- VB Cable Drivers: VB-Audio Virtual Apps

CW TRANSMISSION SETUP

After uploading your Arduino code for CW transmission using the RFM26W module, follow these steps to receive the signal using AirSpy SDR#:

Step 1: Open SDR# (SDRSharp)

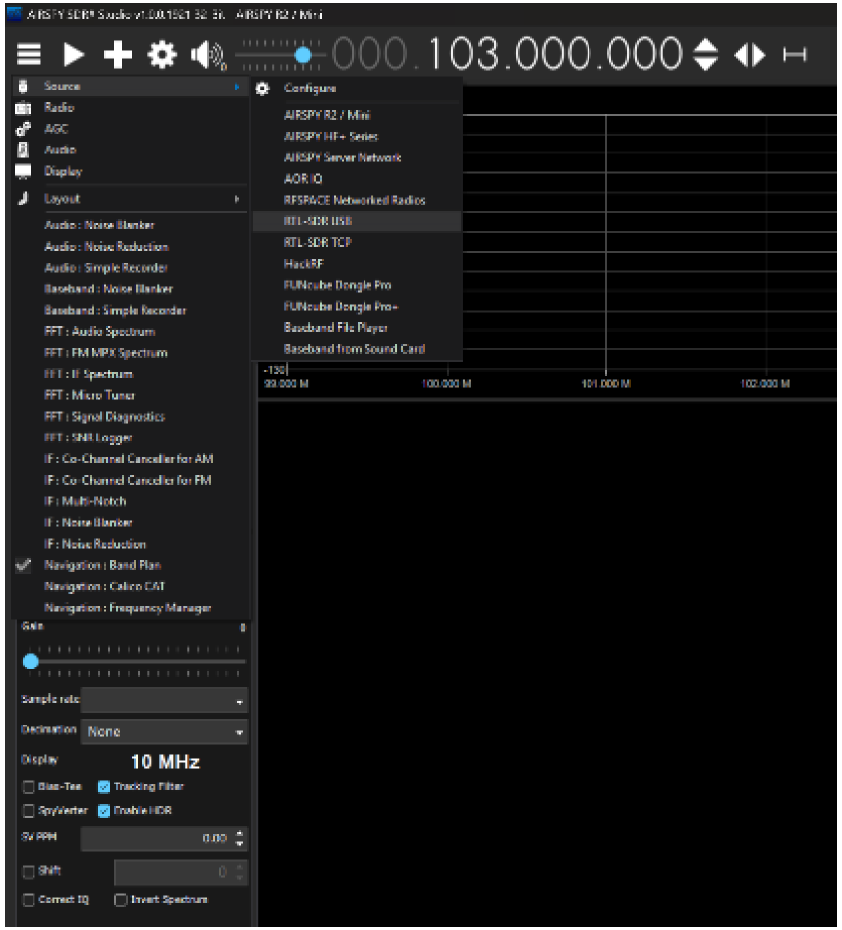
- Launch the SDRSharp software on your computer.



CW TRANSMISSION SETUP

Step 2: Select the RTL-SDR USB as the Source

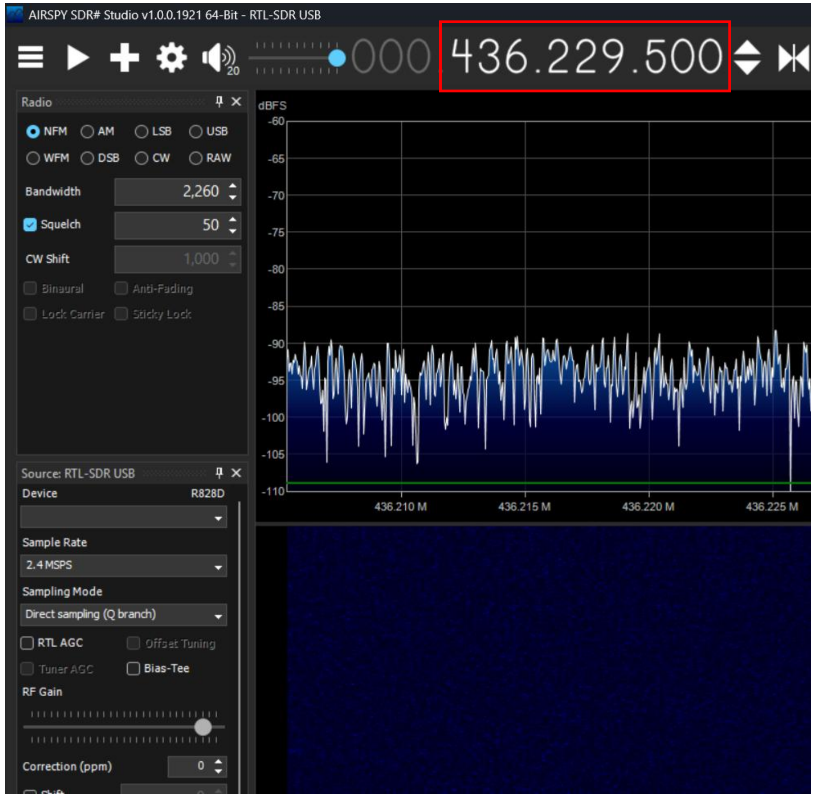
- In the top-left corner of SDR#, click on the three-line menu (☰).
- Navigate to "Source".
- Select "RTL-SDR USB" from the list



CW TRANSMISSION SETUP

Step 3: Set the Frequency

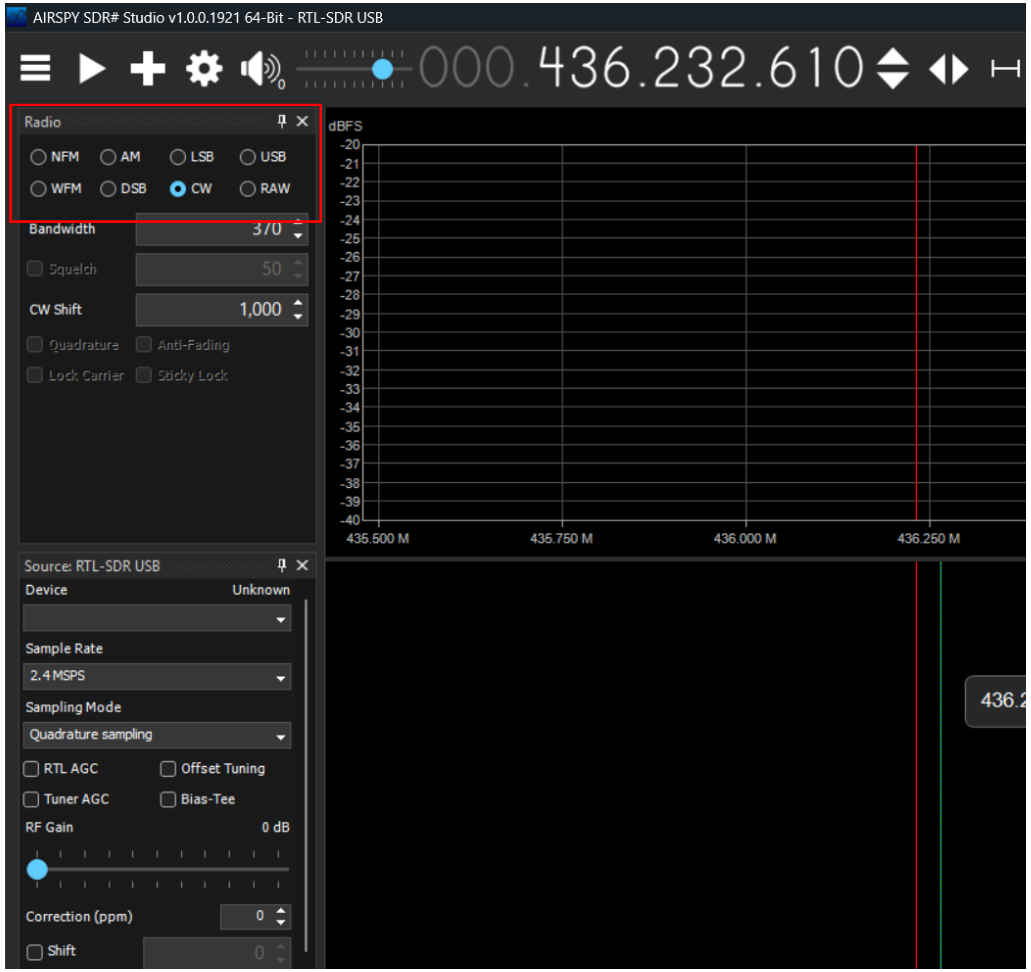
- Locate the frequency input box at the top of SDR#.
- Scroll the mouse wheel to set to the frequency of the RFM26W module(Arduino IDE code default set frequency 436.235MHz).



CW TRANSMISSION SETUP

Step 4: Configure the Demodulation Mode

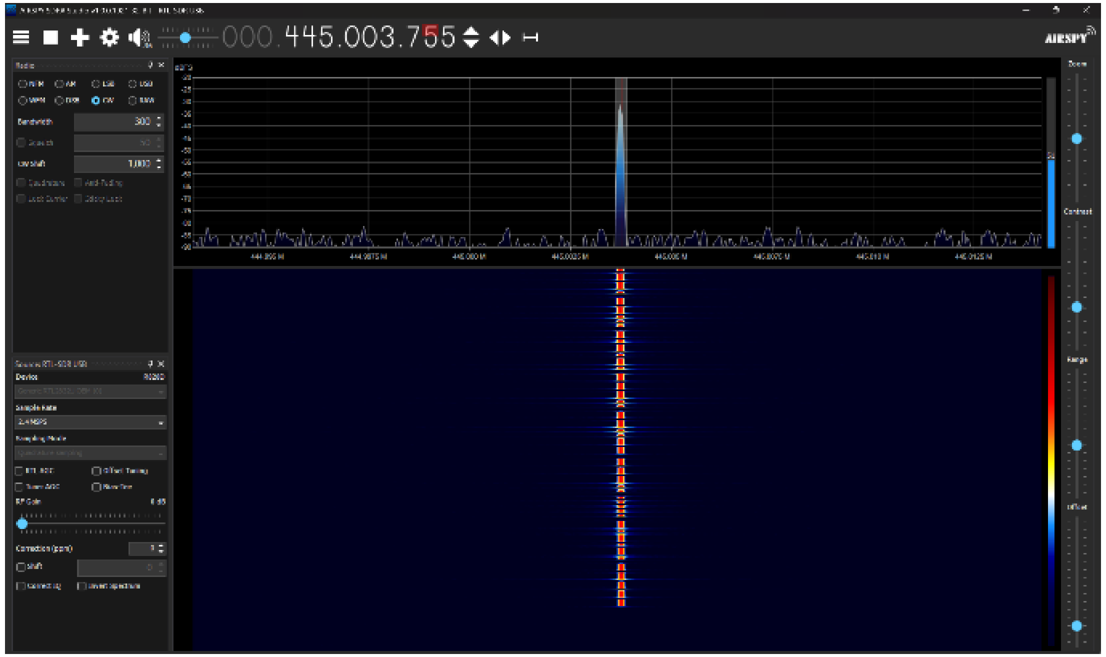
- In the "Radio" section on the left panel, select "CW" (Continuous Wave) as the demodulation mode.



CW TRANSMISSION SETUP

Step 5: Start Receiving the Morse Code Signal

- Click the "Play" button (▶) to start listening.
- If needed, adjust the bandwidth and gain settings for optimal reception.
- You should now hear the Morse Code tones.



CW TRANSMISSION SETUP

Step 7: Decode the Morse Code

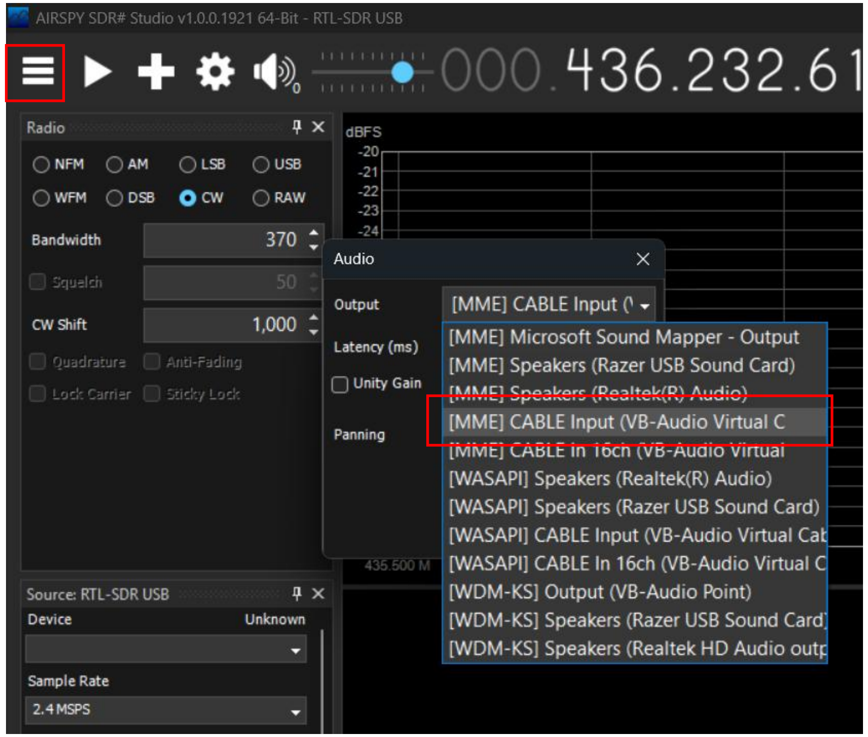
- If you want to decode the Morse Code manually, listen for the short (dots) and long (dashes) pulses.
- Alternatively, you can use Morse Code decoding software such as Morse Decoder apps or CWGet.

CW TRANSMISSION SETUP

Instructions for Decoding Morse Code Received from SDR# Using CWGet

Follow these steps to properly configure SDR# and CWGet to decode Morse code:

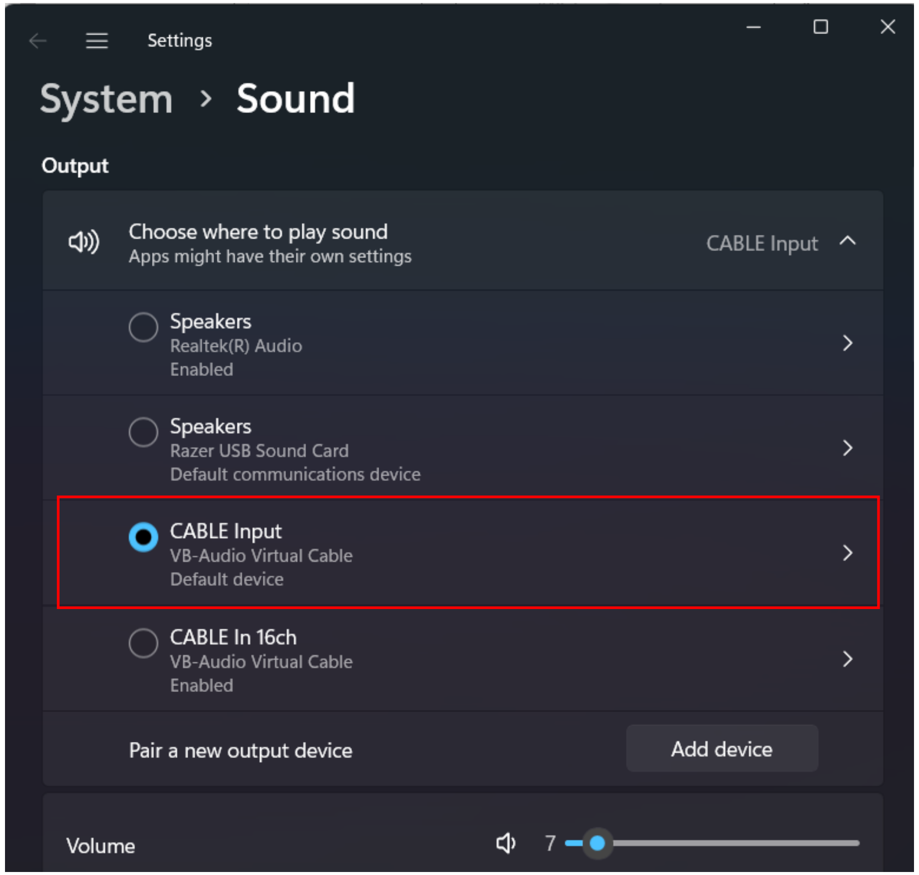
- Open SDR# on your computer.
- Click on the three-line menu (☰) in the top-left corner.
- Navigate to the "Audio" section.
- Locate the Output option and select "Cable Input (VB-Audio Virtual Cable)" from the dropdown menu.



CW TRANSMISSION SETUP

Alternatively, you can set the output sound from your system's speaker settings:

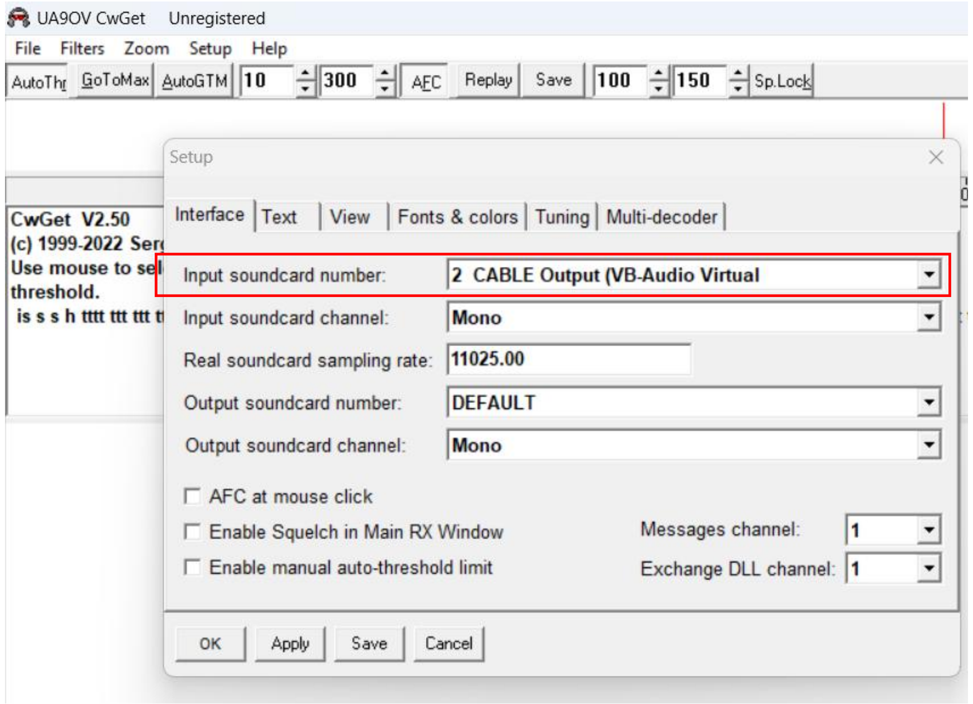
- Right-click on the Speaker icon in the taskbar.
- Select "Sound settings".
- Under Output, choose "Cable Input (VB-Audio Virtual Cable)".



CW TRANSMISSION SETUP

Step 2: Set the Input Sound in CWGet

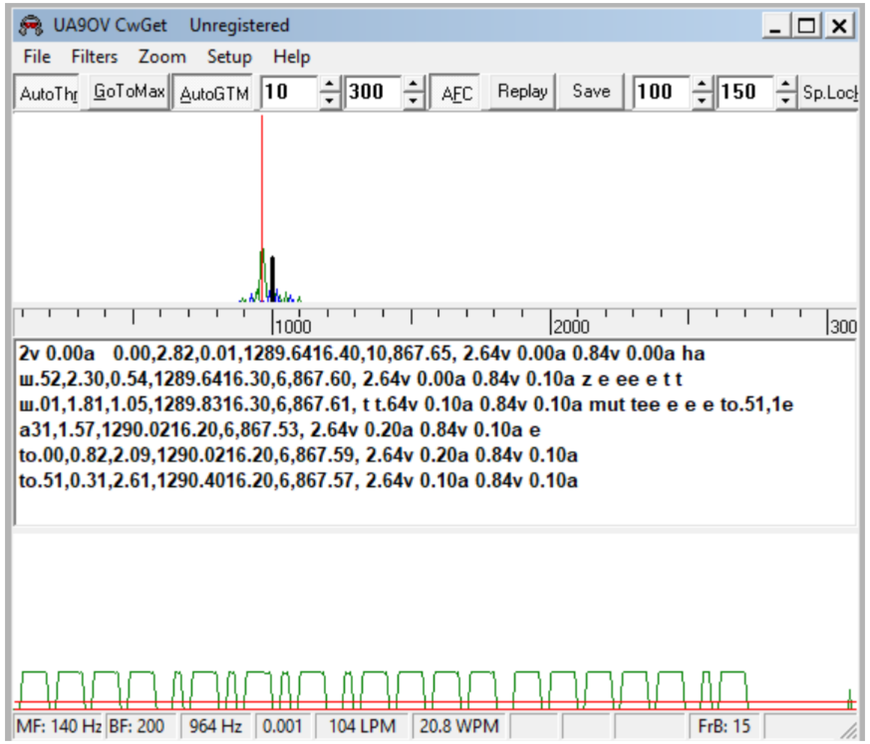
- Open CWGet on your computer.
- Click on "Setup" in the top menu.
- From Setup setting, find "Input Soundcard Number".
- From the dropdown menu, select "Cable Output (VB-Audio Virtual Cable)".



CW TRANSMISSION SETUP

Step 3 : Play the received signal from SDR# and start decoding message.

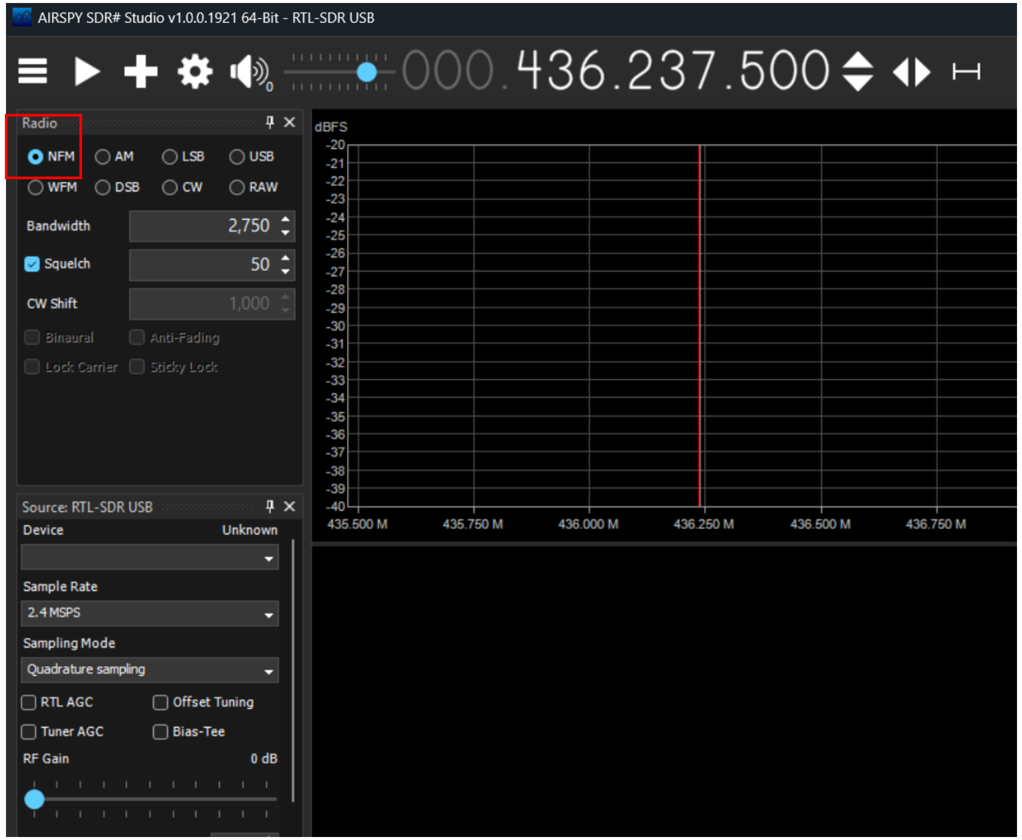
- Go Back to SDR# and Click the "Play" button (▶) to start listening.
- Open CwGet and click on GoToMax button and CWGet will start decoding message from received MorseCode automatically.



GFSK TRANSMISSION SETUP

After uploading your Arduino code for GFSK transmission using the RFM26W module, follow these steps to receive the signal using AirSpy SDR#:

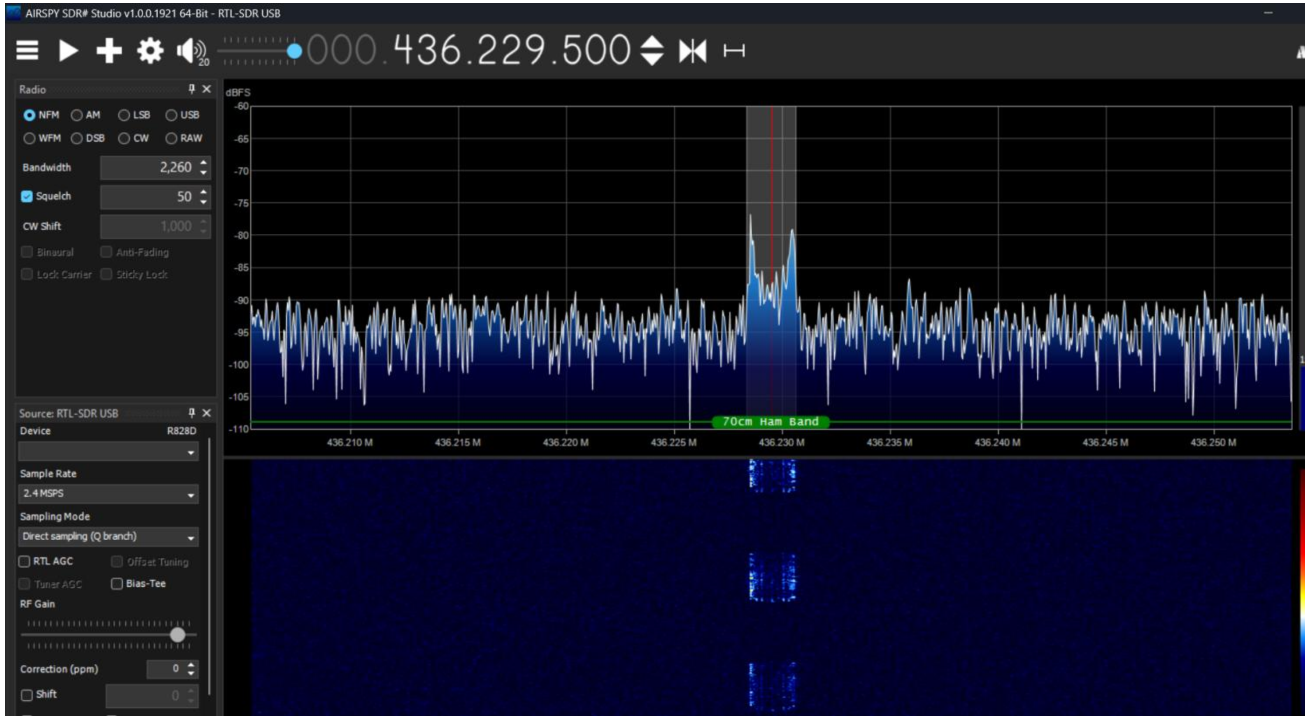
1. Open SDR# software
2. Select the RTL-SDR USB as the Source
3. Set the Frequency (Follow the same steps as CW Transmission Setup upto here)
4. In the "Radio" section on the left panel, select “NFM" (Narrow FM) as the demodulation mode.



GFSK TRANSMISSION SETUP

Start Receiving the Morse Code Signal

- Click the "Play" button (▶️) to start listening.
- If needed, adjust the bandwidth and gain settings for optimal reception.
- You should now hear the GFSK Signal.

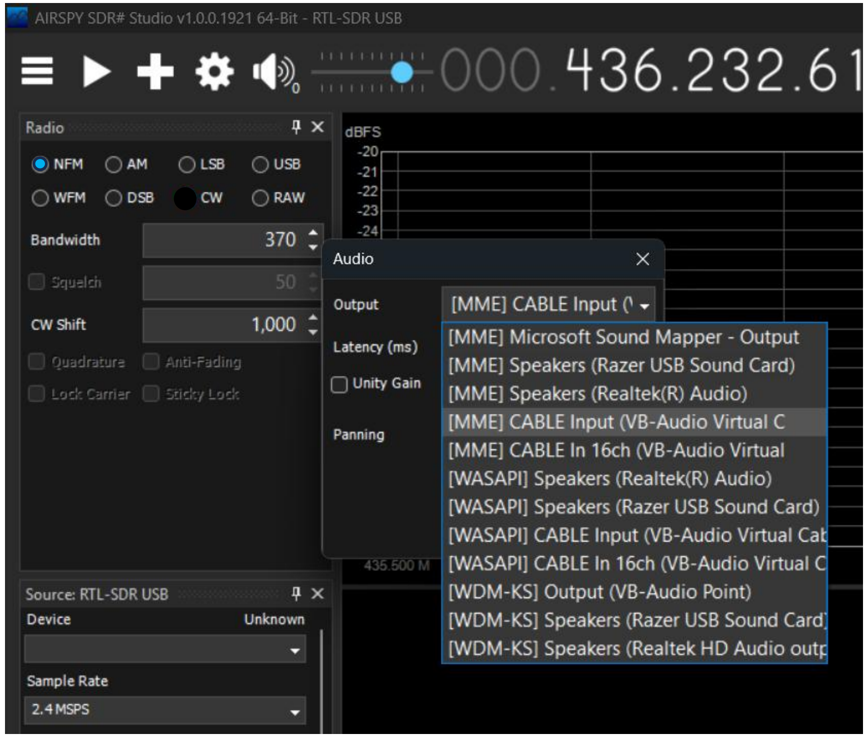


GFSK TRANSMISSION SETUP

Decode the GFSK Signal

- You can use GFSK decoding software such as SoundModem.

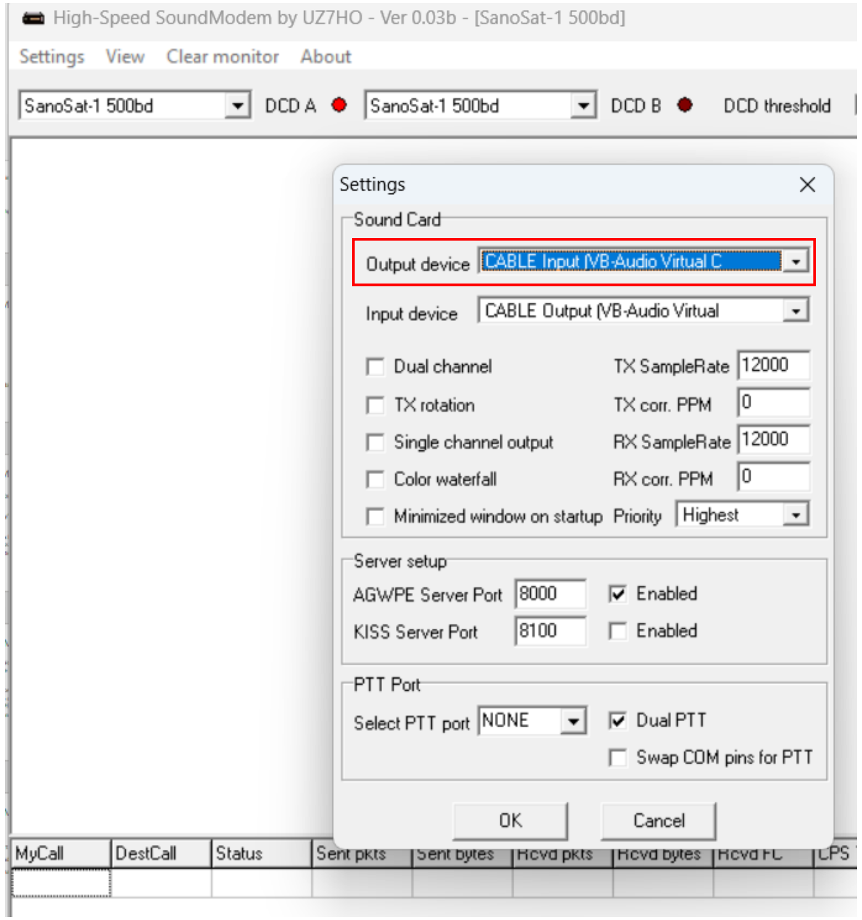
Step1: Locate to Audio - Output option in SDR# and select "Cable Input (VB-Audio Virtual Cable)" from the dropdown menu. (Same as CW Transmission)



GFSK TRANSMISSION SETUP

Step 2: Open SoundModem Configure Output Device

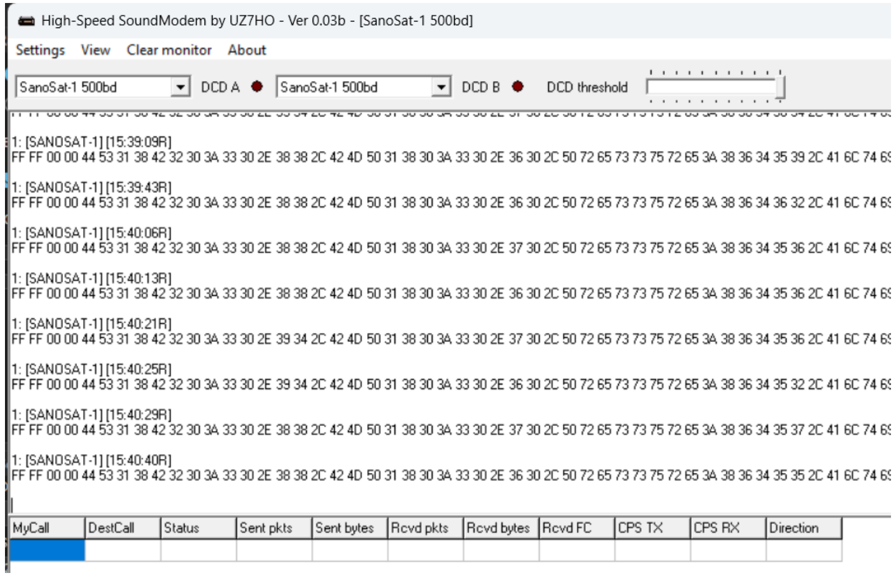
- Open the SoundModem Software
- Navigate to Settings – Devices
- In the Output Devices dropdown menu, select CABLE Input VB Audio Virtual C option and press OK.



GFSK TRANSMISSION SETUP

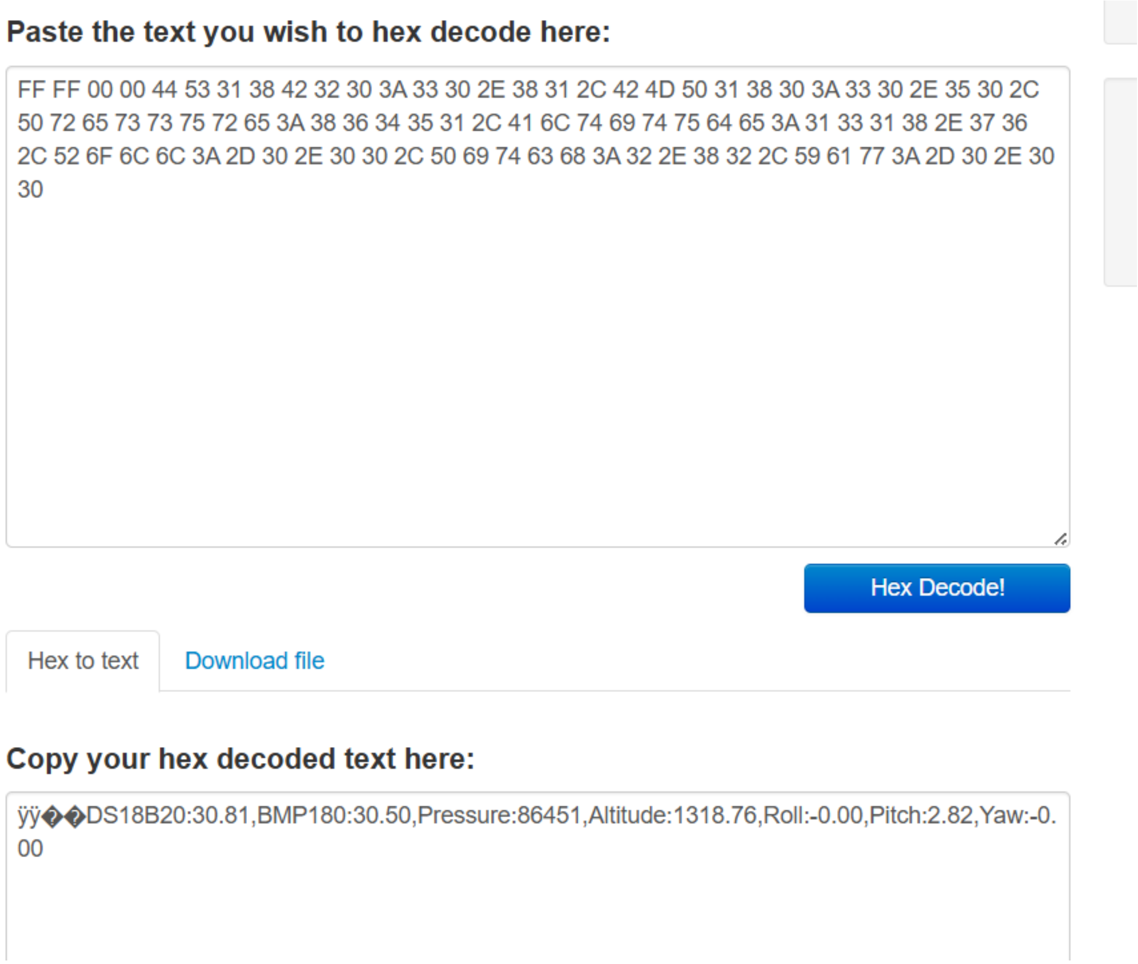
Step 3: Receiving and Decoding GFSK in Hex format

- The SoundModem software should automatically decode the receiving GFSK signal into Hex format in real time.



GFSK TRANSMISSION SETUP

- You can copy the hex output frames and paste them into any hex to text decoders in google for readable format.



GFSK TRANSMISSION SETUP

Important Notes:

- The SoundModem Software only works when the signal is transmitted at 500 Baut rate.
- Hence, ensure GFSK settings match transmitted signal.
- You can check this by searching for rf24.setModemConfig in the Arduino code and see if it is set to RH_RF24::GFSK_Rb0_5Fd1. If not, set it RH_RF24::GFSK_Rb0_5Fd1 to transmit in 500 baud rate.